

Data Report for Stratus 12

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May 12, 2026

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1 Introduction

1.1 Project Overview

About a decade ago the community involved in efforts to improve how well models represented the downward mixing of heat and other properties into the deep ocean recognized the need for additional observations from the deep ocean. Such measurements could be assimilated into ocean models and also used to assess the realism of, for example, storage of heat in the deep ocean. OceanSITES, the international collaborative effort to collect sustained time series observations at open ocean sites under the Global Ocean Observing System (GOOS), launched an effort to coordinate deployment temperature and salinity recorders in the deep ocean on OceanSITES moorings. Funds were raised for a pool of instruments and individual investigators also contributed instruments. The Upper Ocean Processes Group at WHOI joined this effort and committed to deploy pairs of temperature and salinity recorders on its NTAS, Stratus, and WHOTS Ocean Reference Stations. This page is set up to share quality controlled temperature data from the Stratus and NTAS deployments from 2012 onwards. At this point, quality control efforts have produced temperature time series with error estimates. Technical challenges with sensors and quality control on conductivity and salinity continue, and the salinity data are not yet ready for release.

1.2 Deployment Summary

- **Deployment Period:** From 05/17/2012 15:48 to 03/08/2014 17:26
- **Instruments Used:** SBE16 1876 and SBE16 1879
- **Overlap with Next Deployment (Stratus 13):** no

This document details the preprocessing, quality control, and analysis steps for the deep temperature and salinity (T/S) data from Stratus 12 moorings. This report is supplemented by the Python code used in the analysis.

2 Pre-Processing

Detailed steps undertaken during the pre-processing of Stratus 12 data include:

2.1 Data Loading and Initial Processing

- Read .mat files containing SBE16 data using custom functions.
- Convert MATLAB data to xarray datasets for use in Python.
- Process and update metadata.

2.2 Time Correction

- Adjust time data to account for MATLAB's year zero issue.
- Convert time units to 'days since 0001-01-01'.
- Subtract 365 days from converted times to correct for MATLAB's leap year assumption.
- Time base Interpolation (if necessary): Data are interpolated from the initial timing mark to the final timing mark, so this needs to be done before truncation.

Potential timing errors are addressed by referring to the "recover.xls" spreadsheet on buoy5, under the "subsurface" tab, to compare spike start and end times with the apparent time in the instrument data. If the offset is greater than one sample interval (± 5 min for Stratus deployments), it indicates misalignment in time, and the data is linearly interpolated to a new time base to synchronize instruments.

In reviewing the temperature data against recorded spike times for Stratus 12, it is observed that the differences are consistently within 5 minutes. Specifically, for the deployment phase, the significant temperature drop commences at 05/17/2012 15:48, closely aligning with the recorded spike time of 05/17/2012 15:48. Similarly, during the recovery phase, a notable temperature drop starts at 03/08/2014 17:26, which slightly follows the recorded spike time of 03/08/2014 17:26.

Time Correction Decision: Based on these findings, interpolation was not performed for Stratus 12 as the temperature data aligns well with the recorded times, suggesting minimal time offset. See Figs.1 and 2 for details.

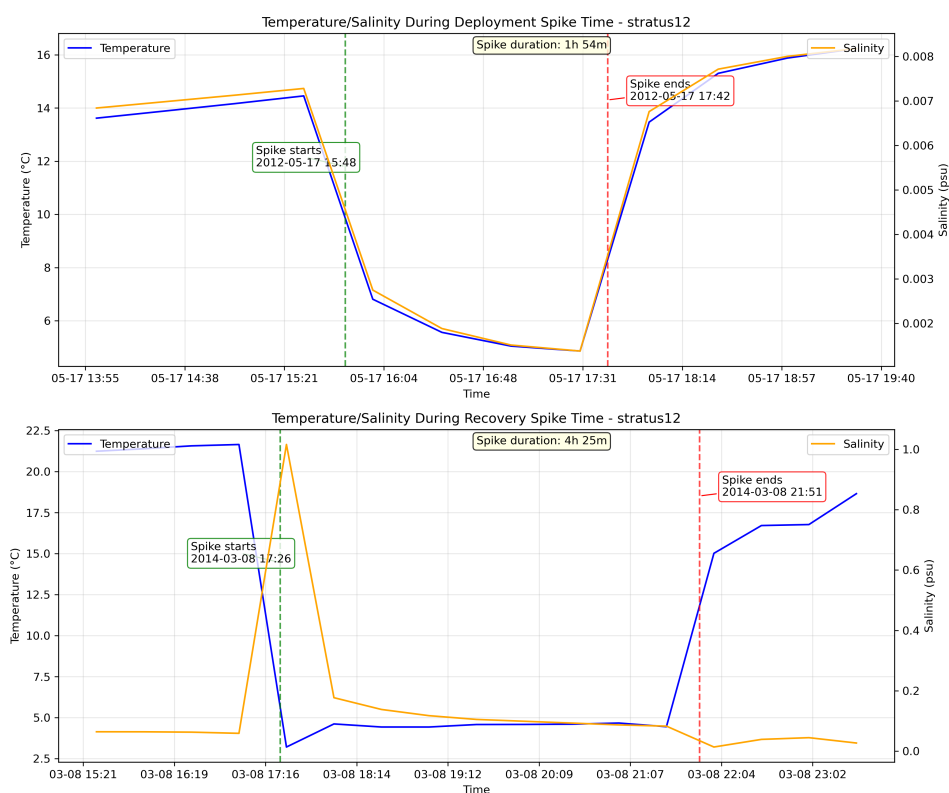


Figure 1: Deployment and recovery spike times for instrument SBE16 1876 of Stratus 12 showing temperature changes during deployment and recovery operations. The vertical dashed lines indicate the recorded spike times (05/17/2012 15:48 for deployment and 03/08/2014 17:26 for recovery).

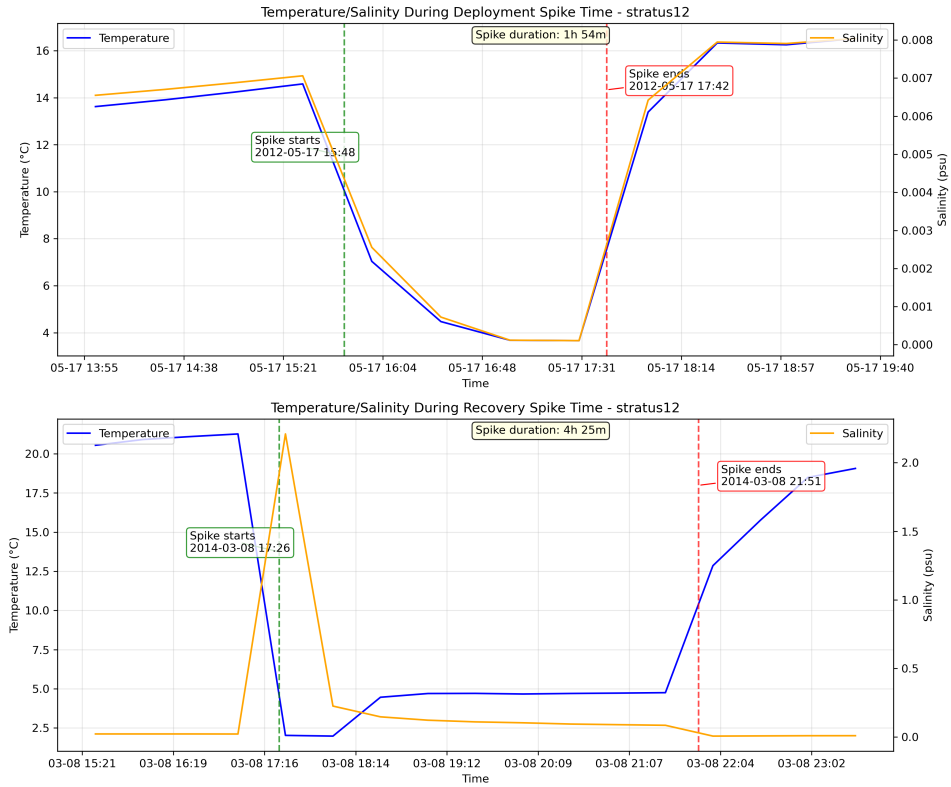


Figure 2: Deployment and recovery spike times for instrument SBE16 1879 of Stratus 12 showing temperature changes during deployment and recovery operations. The vertical dashed lines indicate the recorded spike times (05/17/2012 15:48 for deployment and 03/08/2014 17:26 for recovery).

The deployment and recovery phases are visualized in Figs.3 and 4. The figures show the temperature changes during deployment and recovery operations for both instruments. The vertical dashed lines indicate the anchor over time (anchor dropped) and release fired times (mooring released), which are used to truncate the data.

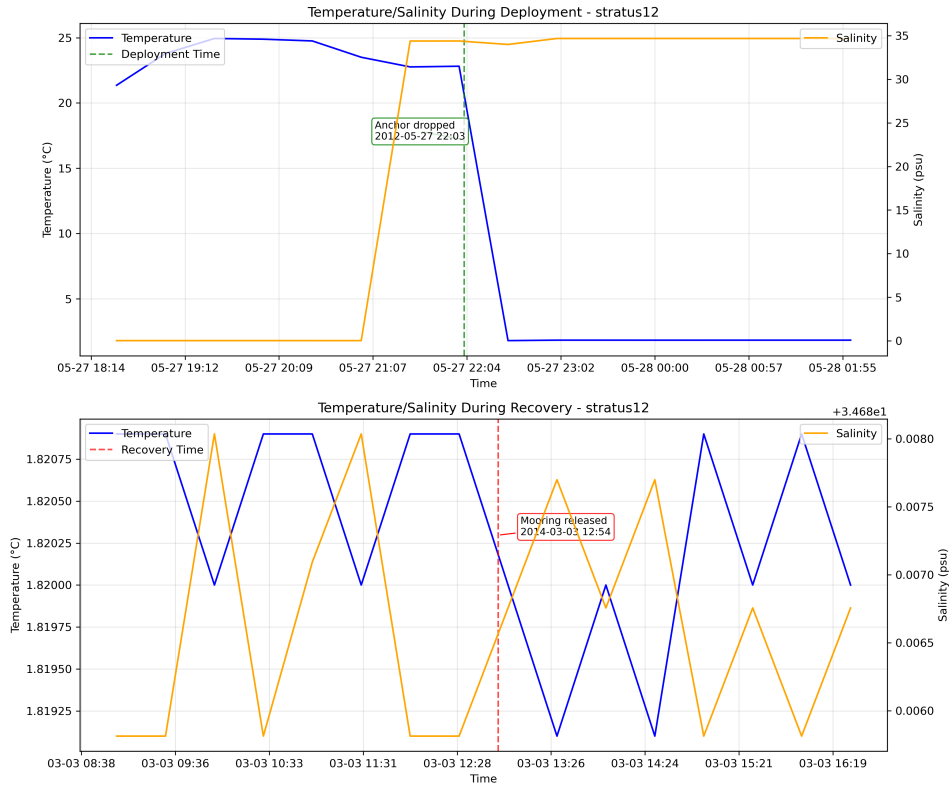


Figure 3: Detailed view of temperature changes during deployment and recovery phases for instrument SBE16 1876 of Stratus 12.

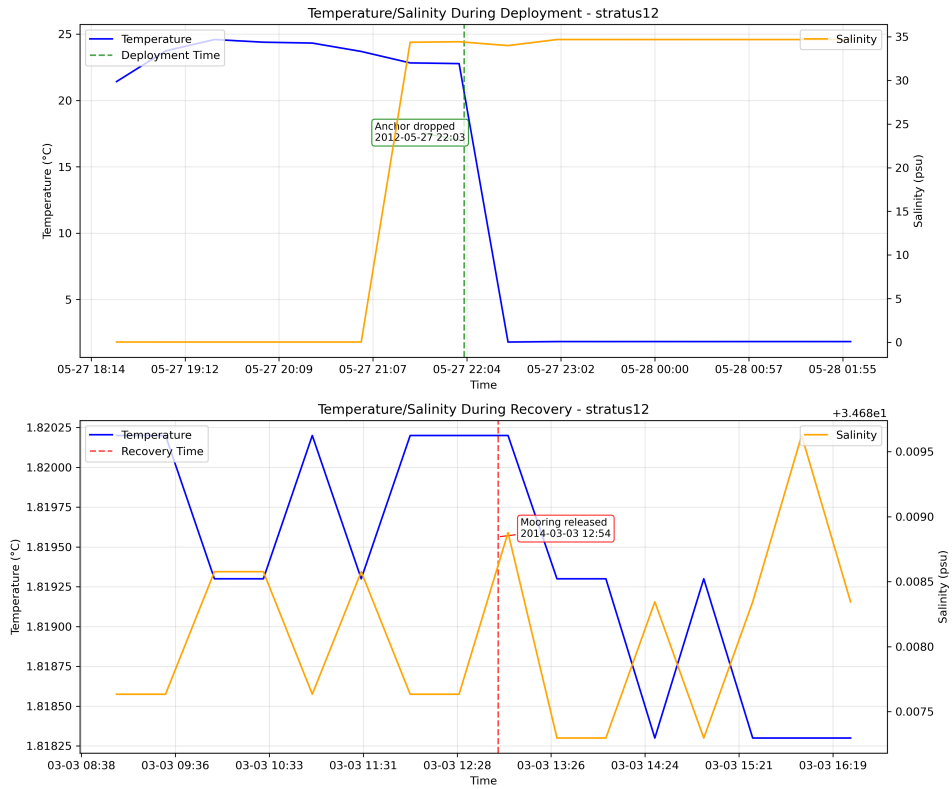


Figure 4: Detailed view of temperature changes during recovery phases for instrument SBE16 1879 of Stratus 12.

3 Truncation and Visualization

3.1 Data Truncation

- Read anchor over time and release fired time from metadata.
- Truncated dataset starts 4 hours after anchor over time and ends at release fired time.
- Plot and inspect temperature time series for both SBE16 instruments.
- Ensure that truncation has successfully excluded data from before deployment and after recovery.
- The time-corrected (if needed) and truncated data files are saved after this stage of processing. These define the "baseline" files for each deployment, before quality control but after time correction and truncation.

The truncated data (Fig.5) shows the expected time window of good data after truncation, starting 4 hours after anchor deployment time and ending at the release fired time.

Comparison within stratus12 Data
Location: (19.9384°S, 85.2927°W) | Depth: 4500m

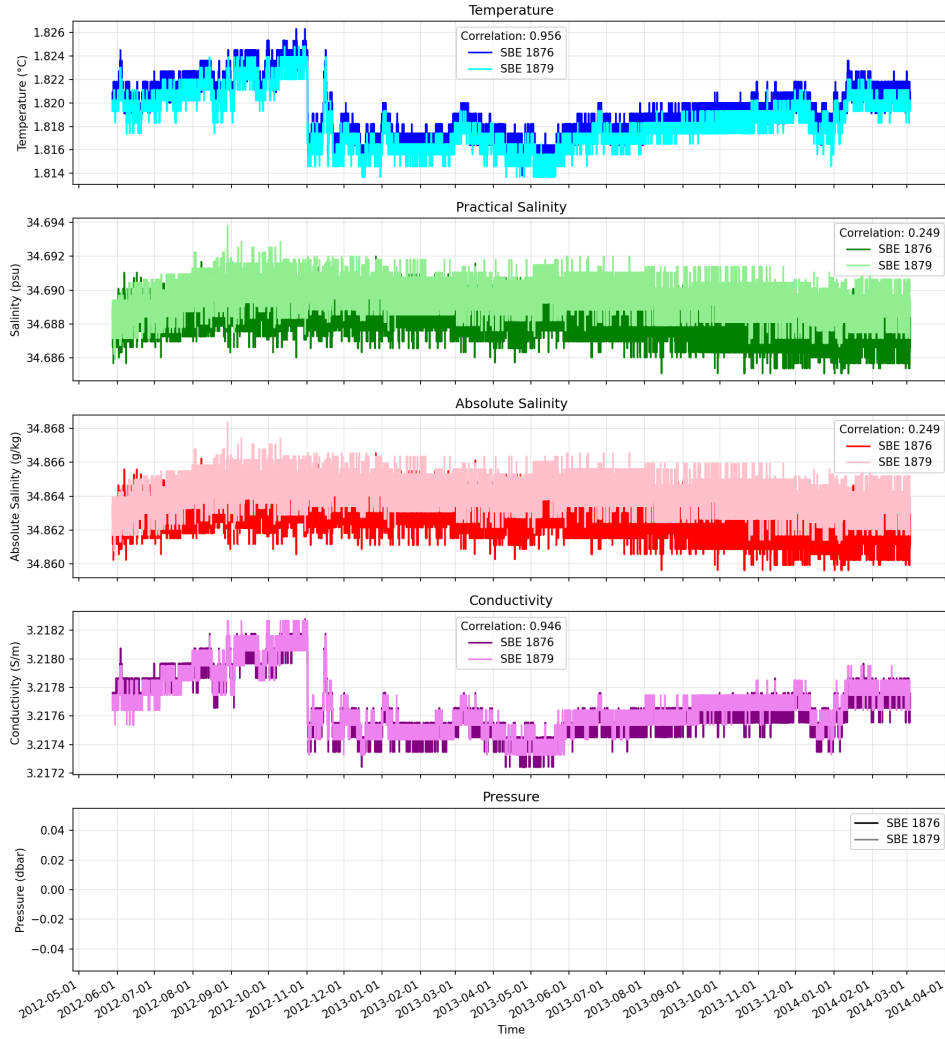


Figure 5: Deployment and recovery timing analysis for Stratus 12, comparing instruments SBE16 1876 and SBE16 1879. This figure shows the expected time window of good data after truncation, starting 4 hours after anchor deployment time and ending at the release fired time. The pressure data is not available for this deployment, so the pressure variable is not shown in this figure.

3.2 Re-calculate Salinity

Pressure data is not available for this deployment, Therefore, we recalculate salinity using a fixed pressure value based on the instrument depth. For Stratus 12, salinity was recalculated using the following procedure:

1. **Pressure Calculation:** Instead of using the potentially drifting pressure sensor data, we calculate a fixed pressure value based on instrument depth:
 - Calculate pressure using the GSW function `p_from_z`, which converts geometric height to pressure

- Use the formula $z = -1 \times \text{instrument_depth}$ (negative because depth is positive downward)
- Use latitude from the anchor survey for the calculation

2. Practical Salinity Calculation:

- Convert conductivity from S/m to mS/cm by multiplying with a factor of 10
- Apply the TEOS-10 (Thermodynamic Equation Of Seawater - 2010) algorithm via `gsw.SP_from_C`
- Use measured temperature and the calculated fixed pressure (rather than varying measured pressure)
- Store the computed values as `sea_water_practical_salinity` with units 'psu'

3. Absolute Salinity Calculation:

- Use the TEOS-10 algorithm `gsw.SA_from_SP` to convert practical salinity to absolute salinity
- Incorporate geographical dependencies using latitude and longitude from the anchor survey
- Store the computed values as `sea_water_absolute_salinity` with units 'g/kg'

4. Post-Processing:

- The original salinity variables (`sal` and `abssal`) are removed from the output datasets

This method ensures consistent salinity calculations and eliminates potential errors from pressure sensor drift, which can be substantial at deep ocean deployments. Using a fixed pressure calculated from the known instrument depth produces more reliable salinity values.

3.3 Data Export

- Ensure all required variables (`sea_water_temperature`, `sea_water_electrical_conductivity`, `sea_water_practical_salinity`, `sea_water_absolute_salinity`, `sea_water_pressure`) exist in the dataset.
- Fill missing data or create non-existent variables with -99999 values and encode them as missing values.
- Carry forward time series for all five variables even if not present, by setting missing values to -99999.
- Save the time-corrected, truncated datasets as NetCDF files for further processing.
- Files are saved in the directory `/data/truncated/Stratus12` as `Stratus12_1876_truncated.nc` and `Stratus12_1879_truncated.nc`.

4 Quality Control and Deployment Catalog

The following quality control measures are implemented:

4.1 Spike Removal

Spike detection and removal are set up in function `src/qc_function.py`. The spike removal function identifies and removes spikes from time series data using a rolling mean and standard deviation. It calculates these statistics within a specified window size around each data point. Points that deviate from the rolling mean by more than 3 times the standard deviation are considered spikes and replaced with missing values (-99999). For Stratus 12 processing, the window size was 12 points and the threshold was 3 times the standard deviation. See Figs.6 and 7 for examples of the original and cleaned plots for temperature, salinity, absolute salinity, conductivity, and pressure for both instruments. These serve as a basis for collaborative review and further data correction.

The spikes removed from the datasets are shown in the table below.

Table 1: Spike removal statistics for Stratus 12

Variable	SN 1879 (%)	SN 1876 (%)
sea_water_absolute_salinity	0.1487	0.0388
sea_water_electrical_conductivity	1.7037	1.7263
sea_water_practical_salinity	0.1843	0.0420
sea_water_pressure	0.0000	0.0000
sea_water_temperature	0.6918	0.3103

4.2 Difference Statistics

When both instruments are available and functional, the mean and standard deviation of the difference between sensors and of individual sensors are computed. These statistics provide benchmarks for accuracy and precision, and are critical for quality control decisions. The difference statistics are computed after spike removal to avoid contamination by known bad points.

The difference statistics are shown in the tables below.

Variable	SN 1879		SN 1876	
	Mean	Std Dev	Mean	Std Dev
sea_water_temperature	1.81831	0.00226	1.81918	0.00226
sea_water_practical_salinity	34.68961	0.00114	34.68826	0.00101
sea_water_absolute_salinity	34.86415	0.00778	34.86281	0.00887
sea_water_electrical_conductivity	3.21768	0.00019	3.21764	0.00020
sea_water_pressure	nan	nan	nan	nan

Table 2: Statistics for individual sensors on stratus 12

Variable	Mean Diff	Std Diff	QC Threshold
sea_water_temperature	-0.00087	0.00320	0.00960
sea_water_practical_salinity	0.00134	0.00152	0.00456
sea_water_absolute_salinity	0.00134	0.01179	0.03538
sea_water_electrical_conductivity	0.00004	0.00028	0.00083
sea_water_pressure	nan	nan	nan

Table 3: Statistics for difference between sensors on stratus 12

Note: These statistics are also stored as metadata in the NetCDF files with the following attributes:

- `std_single_sensor`: Standard deviation of the time series from a single sensor
- `std_sensor_diff`: Standard deviation of the difference between two co-located sensors
- `mean_sensor_diff`: Mean of the difference between two co-located sensors

Important: NaN values that exist in the difference statistics may be due to the conversion process from the original MAT files and do not necessarily indicate data quality issues. The statistics are computed after excluding any NaN values to ensure accuracy.

4.3 Human in the Loop (HITL) Quality Control

A catalog of results for each deployment was created, featuring original and QC plots for all measured variables. See Figs.6 and 7 for examples of the original and cleaned plots for temperature, salinity, absolute salinity, conductivity, and pressure for both instruments. These serve as a basis for collaborative review and further data correction.

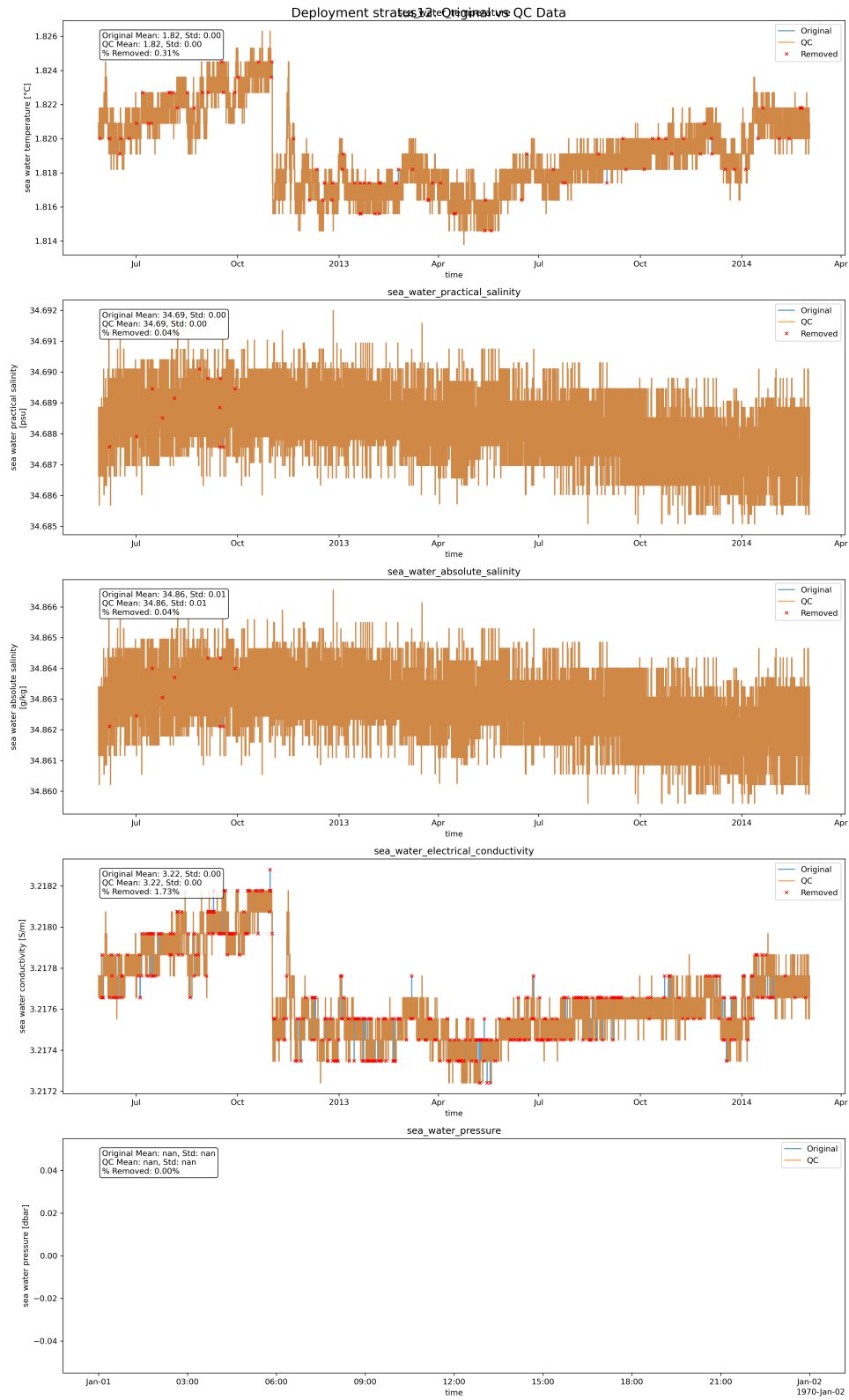


Figure 6: Original and clean plots of temperature, salinity, absolute salinity, conductivity, and pressure for instrument SBE16 1876 of Stratus 12.

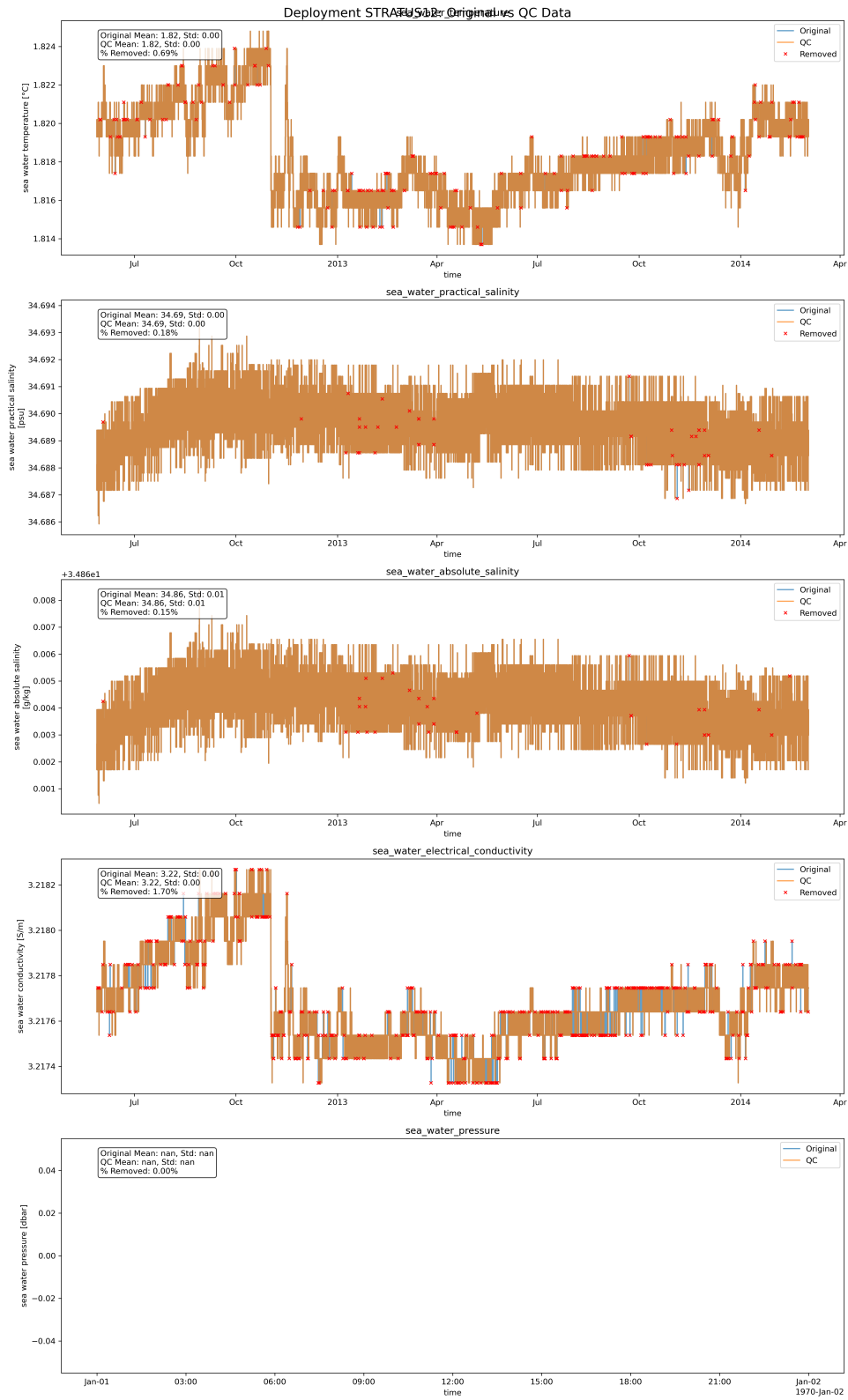


Figure 7: Original and clean plots of temperature, salinity, absolute salinity, conductivity, and pressure for instrument SBE16 1879 of Stratus 12.

4.4 Data Export

- Export the cleaned datasets as NetCDF files for further processing.

- Files are saved in the directory `/data/cleaned/Stratus12` as `Stratus12_1876_cleaned.nc` and `Stratus12_1879_cleaned.nc`
- Each file contains the following variables:
 - `sea_water_temperature` (°C)
 - `sea_water_electrical_conductivity` (S/m)
 - `sea_water_practical_salinity` (PSU)
 - `sea_water_absolute_salinity` (g/kg)
 - `sea_water_pressure` (dbar)

5 Documentation and Archiving

Finalized data sets are documented extensively and prepared for inclusion on the project website. The metadata is updated to be compliant with the NetCDF Climate and Forecast (CF) conventions, and OceanSITES data manual.

6 Conclusion

This report documents the preprocessing, quality control, and analysis steps for the deep temperature and salinity data from Stratus 12. The data has been processed through several stages:

- Time correction and data truncation to isolate the deployment period
- Quality control including spike removal and difference statistics
- Recalculation of salinity using the TEOS-10 algorithm at the instrument depth with observed temperature and conductivity
- Merging of data from two co-located instruments

The resulting cleaned and merged dataset provides a valuable record of deep ocean temperature and salinity at the Stratus site, contributing to long-term climate monitoring efforts. As the first dataset in this processing sequence, it establishes the baseline for future deployments and will serve as a reference point for subsequent analyses.